Consumption of polyunsaturated fatty acids, fish, and nuts and risk of inflammatory disease mortality.

BACKGROUND: n-3 (omega-3) Polyunsaturated fatty acids (PUFAs), fish, and nuts can regulate inflammatory processes and responses. OBJECTIVE: We investigated whether dietary intakes of PUFAs [n-3, n-6 (omega-6), and α-linolenic acid], fish, and nuts were associated with 15-y mortality attributed to noncardiovascular, noncancer inflammatory diseases. DESIGN: The analyses involved 2514 participants aged ≥49 y at baseline. Dietary data were collected by using a semiquantitative food-frequency questionnaire, and PUFA, fish, and nut intakes were calculated. Inflammatory disease mortality was confirmed from the Australian National Death Index. RESULTS: Over 15 y, 214 subjects died of inflammatory diseases. Women in the highest tertiles of total n-3 PUFA intake, compared with those in the lowest tertile of intake at baseline, had a 44% reduced risk of inflammatory disease mortality (P for trend = 0.03). This association was not observed in men. In both men and women, each 1-SD increase in energy-adjusted intake of α-linolenic acid was inversely associated with inflammatory mortality (hazard ratio: 0.83; 95% CI: 0.71, 0.98). Subjects in the second and third tertiles of nut consumption had a 51% and 32% reduced risk of inflammatory disease mortality, respectively, compared with those in the first tertile (reference). Dietary intakes of long-chain n-3 and n-6 PUFAs and fish were not associated with inflammatory disease mortality. CONCLUSIONS: We report on a novel link between dietary intake of total n-3 PUFA and risk of inflammatory disease mortality in older women. Furthermore, our data indicate a protective role of nuts, but not fish, against inflammatory disease mortality.